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EARLY PREDICTION OF TYPE2 DIABETES USING SUPERVISED MACHINE LEARNING ALGORITHM

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ABSTRACT

Diabetes is a common chronic health condition with significant implications on a global scale. Early and accurate prediction of the health condition of the patient suffering from diabetes is extremely beneficial in enhancing the health conditions of the patient. Blood glucose levels are significant factors in the determination of the presence of the health condition known as diabetes, which is a type of metabolic disorder. Machine learning algorithm performance is significant in enhancing the diagnosis and prognosis of various health conditions. The classification algorithms, namely Random Forest (RF), Decision Tree (DT), Naïve Bayes (NB), and Logistic Regression (LR), were used for the experiment. The performance measures like area under the curve, F1 score, recall, accuracy, and precision are used for testing the performance of the algorithm. The results of this investigation showed that RF performs better than the other models, DT, NB, and LR, in terms of accuracy (77.49%), precision (68.97%), and sensitivity (59.46%) and AUC-ROC of 83.8. Diabetes is a global problem that impacts individuals everywhere. Predictive models based on machine learning can help detect diabetes earlier before it becomes a serious condition.

Keywords: Machine Learning, Diabetes, Random Forest, Accuracy, Decision tree, Accuracy

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1. INTRODUCTION

Diabetes mellitus is a chronic metabolic disorder that occurs due to high blood glucose levels. If the glucose is not absorbed by the cells of the body properly, it results in the accumulation of glucose in the blood, which may cause several issues related to the functioning of the organs. Diabetes mellitus may increase the risk of early death and may also cause issues in several aspects of the body. It is a serious health concern that has been rising every year. The past few decades have seen an increase in the number of cases of diabetes mellitus. It is therefore imperative and highly beneficial to carry out research work on the speedy and accurate diagnosis of diabetes.

Diabetes is a lifelong health issue that affects human body and there are several types of diabetes, each with its own causes and characteristics. The main ones are Type 1, where the body's immune system attacks and stops it from making insulin; Type 2, which happens when the body doesn't respond well to insulin or doesn't make enough; and gestational diabetes, which shows up only during pregnancy. Knowing about these types helps with better spotting, treating, and handling the disease.

According to the International Diabetes Federation (IDF) Diabetes Atlas, 2024, an estimated 589 million adults aged 20-79 years, which is about 11.11% of the global adult population, were living with diabetes worldwide, and this is expected to increase to approximately 853 million adults by 2050. In India, which has the second highest number of adults with diabetes, about 89.8 million adults aged 20-79 years had diabetes in 2024, and this is expected to rise to 156.7 million by 2050.

Millions of people worldwide have been afflicted with diabetes, a diverse and non-communicable disease, at an alarming rate. Patients with diabetes and those at risk for developing the disease often exhibit the following early symptoms and indicators: increased thirst, weariness, weight gain, lightheadedness, discolored skin, and sexual deficiency. Human organs such as the brain, nerve damage, heart, kidney, eyes, skin, and so on can all be negatively

impacted by diabetes mellitus. Diabetes is causing a patient's pancreatic beta cells to be pathologically destroyed [1].

2. LITERATURE REVIEW

The machine learning approach for diabetes prediction that combined the XGBoost hybrid fusion method with the Random Forest (RF) algorithm. The study emphasized the significance of feature selection in improving the performance of the fusion model by identifying the most relevant factors for diabetes prediction. The authors showed the effectiveness of optimizing the model using feature selection, which can lead to better patient outcomes and a personalized approach to managing diabetes [2].

In this experiment, a collective methodology was used, including state-of-the-art machine learning models such as logistic regression, Random Forest, SVM, naïve Bayes, AdaBoost, GB, CatBoost, XGBoost, and bagging. Performance metrics used included F1 score, recall, precision, and accuracy metrics. Based on the experiment using the PIMA diabetes dataset, the proposed method showed a precision, accuracy, F1 score, and recall of 73.48%, 79.04%, 80.6%, and 71.45%, respectively [3].

The study proposed a mobile-based model that utilizes different machine learning techniques to predict an individual's likelihood of acquiring diabetes. The proposed system was assessed using a private dataset and (PIDD). An ensemble models, was conducted to assess their effectiveness in diabetes prediction. The results showed that XGBoost with SMOTE oversampling yielded better results compared to other models. The proposed system attained an accuracy of 97.4%, F1-score of 0.95, and AUC of 0.87 using a private dataset. The proposed system also attained an accuracy of 83.1%, F1-score of 0.76, and AUC of 0.85 using a combined dataset [4].

A study titled “An Explainable Ensemble Learning Model for Diabetes Prediction Using Random Forest, XGBoost, and LightGBM Algorithms” proposed a model that utilized different models such as Random Forest, XGBoost, and LightGBM to develop a predictive model for diabetes. The proposed system used SHAP to identify the most significant predictors as well as rank feature importance. The LightGBM model with SHAP-based feature selection improves the model's performance as well as its ability to achieve an accuracy of 80-82% with better interpretability [5].

To assist diabetes researchers, the above reviews have evaluated the performance of SVM and DT-RF models using various parameters such as accuracy, precision, recall, F1, ROC, AUC, and speed. The multi-modal deep learning model using attention mechanisms and

temporal data fusion for the diagnosis of type 2 diabetes-related warnings and complications reported 94.7% accuracy [7]. According to the study, the feature selection using SHAP increased the classification accuracy. Moreover, overfitting was reduced. The performance of the LightGBM model was better compared to other machine learning techniques. The model reported an accuracy of nearly 82%, with an AUC score of 0.89[8].

RF attained an accuracy of 80%, precision of 82%, and sensitivity of 88% in early diagnosis by applying PCA and visualization techniques [9].

Modak and Jha in their study published in 2024 used an ensemble technique, which yielded outstanding results by obtaining an accuracy of 95.4% and an AUC-ROC of 0.99 by CatBoost. The hybrid models combined CNN-KNN for 95% accuracy in mixed data sets for early interventions [11]. RF and XGBoost were used in testing for up to 99.27% accuracy in various sets of data, beating prior predictions. Overall, the studies reflect the evolution of ML towards precision in diabetes prevention and management [12].

3. RESEARCH METHODOLOGY

This section will describe the workflow of the proposed framework for the prediction of diabetes. The steps involved in the workflow include dataset preparation, data preprocessing,

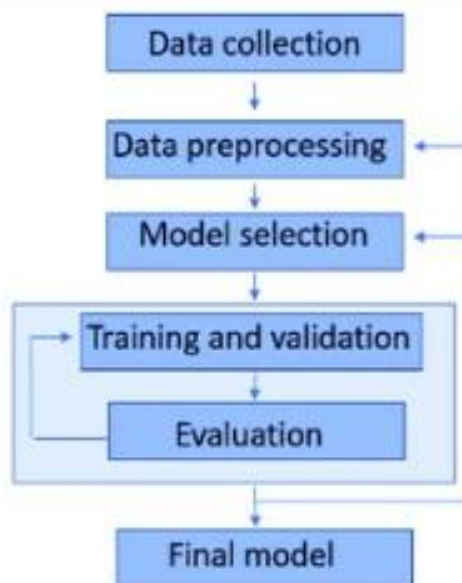


Fig.1. The Proposed Machine Learning Model

3.1 Data Collection Procedures

The data set used for this study is Pima Indian Diabetes data set available in UCI repository of machine learning data. This data set comprises 768 sample female patients from Arizona, USA, who were tested for diabetes. In this data set, there are 8 attributes in total,

representing medical diagnosis criteria. In addition, there is one target class, representing the status of each tested individual. In this data set, there is a total of 268 tested positive instances and 500 tested negative cases. The dataset includes following attributes.

Table 1: Description of the PIMA Indian Dataset Attributes

Attributes	Inference
Insulins	Amount of insulin in a 2 h serum test
Number of pregnancies	Number of times the person has been pregnant
Glucose concentration	Blood glucose level on testing
Age	Age of the person
BMI	Body mass index
Skin thickness	Skin fold thickness of the triceps
Diastolic blood pressure	Diastolic blood pressure
Diabetic pedigree function	DPF calculates a score that represents the likelihood of diabetes influenced by hereditary factors
Outcome	0 for non-diabetic and 1 for diabetic

3.2 Data Preprocessing

The following procedures are involved in data preprocessing:

Data Collection: Data collection entails obtaining the information needed for the research. This study has used PIMA datasets as shown in Table 1

Data Preparation and Cleaning. This entails finding and fixing mistakes or discrepancies in the data, like duplication, outliers, and missing numbers. The mean value of the corresponding attribute has been imputed to handle the missing values in order to improve the model's performance.

Handling outliers: There are extreme values which are different from the norm and are also inconsistent with other values in the data set. The interquartile range (IQR) method can be used to detect outliers in a data set. The IQR, or the difference between the first quartile (Q1) and third quartile (Q3), has to be computed first. Outliers are those values that are less than $Q1 - 1.5 \times IQR$ or more than $Q3 + 1.5 \times IQR$.

Label Encoding: The process of converting text or category value labels into a numerical representation that machine learning algorithms can comprehend is known as label encoding. The categorical values of gender status, for instance, can be changed from "Male" to "1" and "female" to "0." **Standardization/ Normalization.** Normalization in machine learning

is a data scaling technique where numerical features are transformed to fit within a specific range usually $[0, 1]$ or $[-1, 1]$.

Feature Extraction: For improving the performance of ML models, numerical values such as glucose and BMI were standardized to ensure that all attributes are on the same scale. Feature normalization or standardization helps to improve the performance and accuracy of ML models. Principal Component Analysis (PCA) is used to reduce the dimensionality of the data by finding the most important components of the data set.

Splitting Datasets: For training ML models, the dataset is split into a Training Set, i.e., 80% of the total dataset. The testing of ML models is done by using unseen data, i.e., a Testing Set, which is 20% of the total dataset.



Fig. 2. After Preprocessing Dataset on PIMA Data

3.3 Machine Learning Model

A classifier that translates the input variables to target classes based on the training data. The main goal of employing a classifier during the entire investigation is to predict whether the patient has a diabetes or not. The classification model is used to predict the test data set.

Naïve Bayes

Naive Bayes is a classification algorithm based on probabilistic principles and depends on Bayes' theorem for its operation. Naive Bayes operates on the principle that all features are independent of one another, which earns it the 'naive' prefix. Naive Bayes calculates the probabilities for instances belonging to a particular class based on the probabilities for its features, assuming that none of its features are dependent on another [14].

Random Forest

Random Forest is an ensemble learning algorithm used for both classification and regression tasks. It builds multiple decision trees during training and combines their outputs to improve predictive accuracy and reduce overfitting. In order to produce a more reliable and accurate final forecast, it first trains a 'forest' of multiple decision trees, and then it makes a forecast using a combination of individual predictions made by each tree [15].

Decision Tree

DT is a model that is created using a flowchart method with a tree-like structure. It can be used to perform different operations such as classification or regression. The process involves partitioning the given data into smaller subsets depending on the feature values, hence creating a DT model. The dataset is split into subsets based on a feature that has maximum information or minimum uncertainty. This process continues until a predetermined depth for the tree is achieved. The leaves of the tree are predictions [16].

Logistic Regression

Logistic regression is a statistical technique used to describe the relationship that exists between a dependent variable and one or more independent variables. This technique has often been used in machine learning, for example, classification, to determine how a dependent variable is affected by independent variables and to compute the probability of a result [17].

K-fold cross-validation

This approach is particularly useful when the dataset is limited. However, an inappropriate choice of k can lead to misleading estimates of model performance, such as high variability in accuracy. This experiment divides the data into ten subsets and uses a 10-fold cross-validation technique.

Confusion Matrix

A machine learning algorithm's performance summary on a specific set of test data is called a confusion matrix. When evaluating classification models that is, models that make predictions based on classifications the method works well.

The number of accurate and inaccurate predictions is shown in a confusion matrix: True Positive (TP): Correctly predicted positive cases.

- False Positive (FP): Incorrectly predicted positive cases (actually negative).
- True Negative (TN): Correctly predicted negative cases.
- False Negative (FN): Incorrectly predicted negative cases (actually positive).

Performance Metrics

Accuracy (ACC). The ratio of all accurate predictions to all instances of data is known as accuracy. The total number of patients whose diabetic status was accurately predicted as either diabetic or non-diabetic is referred to as accuracy in this research.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

Precision

Precision is the ratio of the total number of accurate positive predictions to the total number of positive predictions. This is also referred to as the measure of confidence in the predictions.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

Recall (Sensitivity): Recall is calculated by dividing the accurate number of positive predictions by the total number of positive predictions. Recall is a measure of the number of relevant patients that have been correctly detected, i it is also called Sensitivity or true positive rate (TPR).

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

F1 score: The F1-score is the harmonic mean of precision and recall, providing a balance between these two metrics. It is calculated as twice the product of precision and recall divided by their sum. The F1-score gives an overall measure of a model's prediction quality.

$$\text{F1 score} = 2 * (\text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})$$

ROC/AUC Curve: It indicates how effectively the model can distinguish between different classes. If the value of AUC is high, the model predicts classes indicated by 0 as value 0 and classes indicated by 1 as value 1. By analogy, when the value of the AUC is high, the model is more efficient and therefore we can distinguish patients with disease and without disease.

4.RESULTS

The dataset is used to predict whether a person suffers from diabetes or not. The result, which represents whether a patient is suffering from diabetes or not, is shown in the pie chart in terms of percentage. According to the results, 500 non-diabetic cases, making up 65.1% of the total subjects, and 268 diabetic cases and, making up 34.9% of the total count as illustrated in a pie chart representation and distribution as outcome .

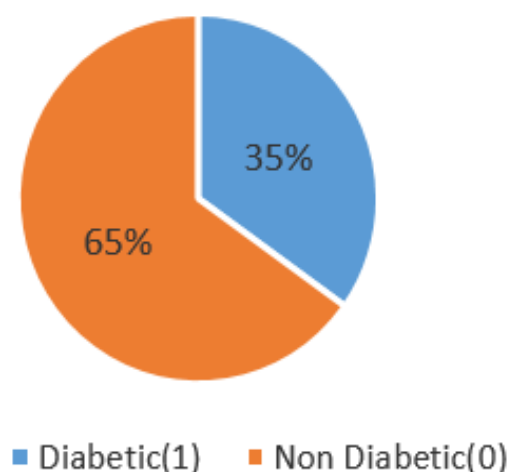


Fig. 3. The distribution of the PIMA Dataset's Outcome Variable

In the current study, the focus was on the analysis, the independent feature was used to foresee the being or absence of diabetes. The classification model LR, RF, NB, and DT were used in the current study, and the results were obtained using the 10-fold cross-validation method. The results of the models are discussed in the following sections. Table 2 shows the statistical summary of the dataset's attributes.

Table 2: Statistics of PIMA Indian Dataset

Statistics	Pregnancies	Glucose	Blood Pressure	Skin Thickness	Insulin	BMI	DPF	Age
Count	768	768	768	768	768	768	768	768
Mean	3.84	121.60	72.20	26.60	118.68	32.4	0.47	33.24
Std	3.36	30.40	12.10	9.60	93.08	6.80	0.33	11.76
Min	0.00	44.00	24.00	7.00	14.00	18.20	0.07	21.00
Max	17.00	199.00	122.00	99.00	846.00	67.10	2.42	81.00

Table 3: Model Performance of Classification Model

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
DT	74.46	61.11	54.05	57.38	75.66
RF	77.49	68.97	59.46	63.86	83.86
LR	77.06	68.57	58.11	62.90	82.07
NB	76.19	66.67	58.11	62.07	82.64

The overall results obtained from all developed models on the prediction of diabetes are presented in Table 3. From all models, it can be seen that the highest accuracy, 77.49%, was obtained from the Random Forest (RF) model, followed by Logistic Regression (LR) with an accuracy of 77.06%, then Naïve Bayes with an accuracy of 76.19%, and finally, the lowest accuracy, 74.46%, was obtained from the Decision Tree (DT) model. The Random Forest model was found to be the best in all evaluation metrics.

The visual representation of performance of model is given figure 4 and performance of model in terms of accuracy is presented in figure 5.

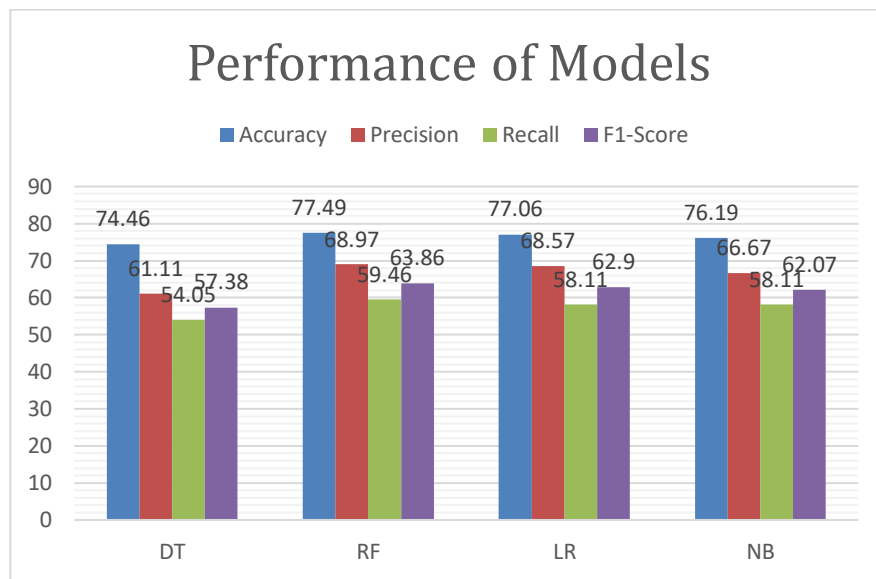


Fig 4. Performance of Models on PIMA Dataset

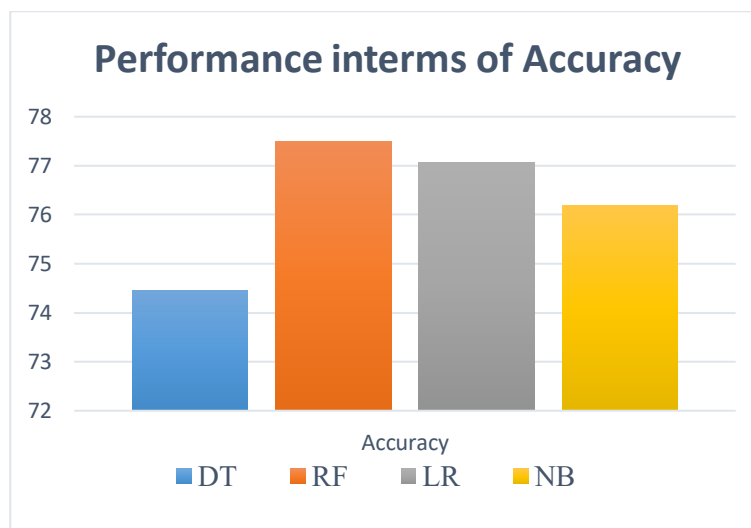


Fig. 5. Performance in terms of Accuracy

5.DISCUSSION

Comparisons of Performance Analysis

From the comparison of all the developed models, it is clear that the best accuracy is achieved by RF with an accuracy of 77.49%, and it is the most successful in identifying positive cases with a sensitivity of 59.46%, a good level of precision with 8268.97%, and an F-score of 63.86%.

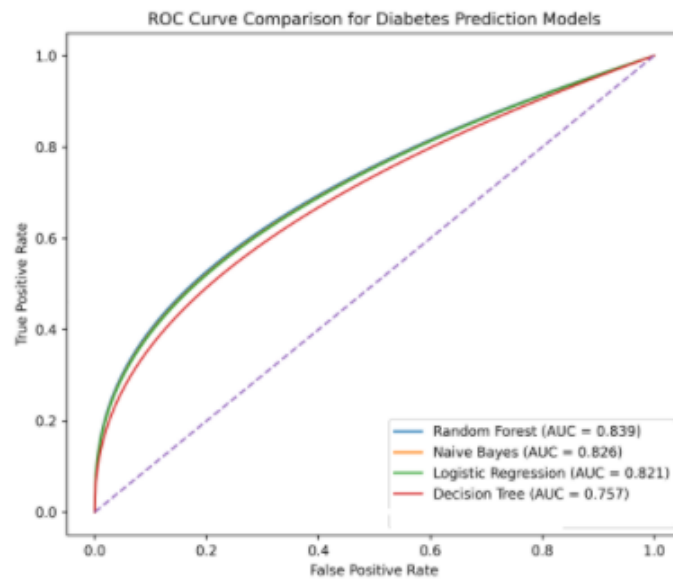


Figure 6. ROC curve Comparison for Diabetes Prediction

The ROC–AUC analysis demonstrates that the Random Forest model achieves the highest performance with an AUC of 0.8386, indicating superior ability to differentiate between diabetic and non-diabetic cases. Naïve Bayes and Logistic Regression display comparable performance, with AUC values exceeding 0.82, reflecting reliable and stable classification behavior.

Analysis of Correlation

In order to find the relationship between the different attributes of the dataset, correlation analysis was done, and a heatmap was produced. Figure 6 shows the correlation between a 0 to 1.0 positive correlation and a 0 to -1 negative correlation in terms of the color format.

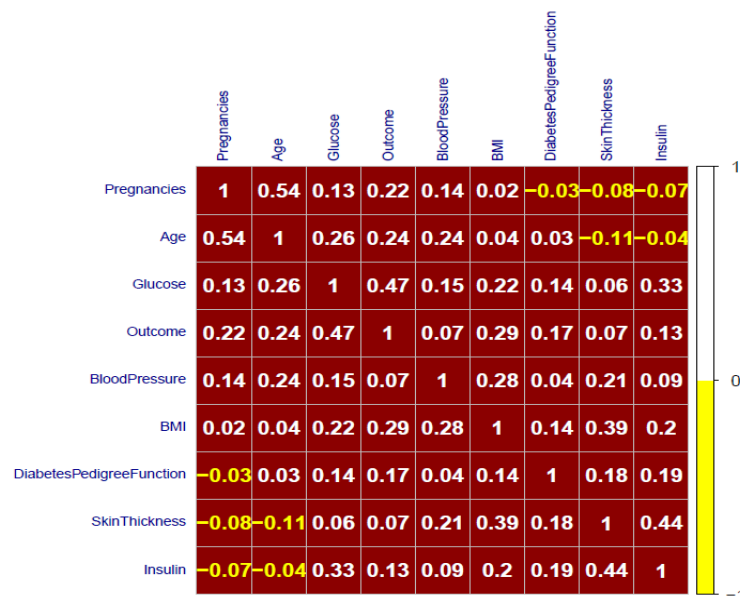


Fig.7. Correlation between attributes

The values of the correlations appear on the heatmap in the colors of yellow, brown, white, red, and pink. High positive correlations appear in dark brown to black, moderate positive correlations appear in red and pink, weak and nearly zero correlations appear in pale yellow to white, and negative correlations appear in faint brown. The study, which had a mean age of 33, found that there was a weak to moderate negative correlation in the levels of insulin, skin thickness, and age.

6.CONCLUSION

Diabetes is characterized by elevated blood sugar levels and can lead to severe health complications that reduce life expectancy. In this study, machine learning models have been created using the PIMA dataset. Heatmap analysis has been used to determine the relationship between the features. The programming language Python has been used to train the model using various parameters such as accuracy, precision, error rate, and ROC. From the study, it is evident that RF is superior to the other models, namely DT, NB, and LR, in terms of accuracy (77.49%), precision (68.97%), and sensitivity (59.46%). In addition, it is evident that there is a weak to moderate negative correlation between insulin, skin thickness, and age, providing more information on the relationships between the features, which are important in the diagnosis of diabetes. Diabetes mellitus is a non-communicable disease, which means it is a chronic disease with no complete cure. There are a number of risk factors, like a poor lifestyle, obesity, etc., which might be a major problem for a diabetic patient. Preventive measures and awareness are always helpful in reducing the risk of this disease occurring in an individual's body. In the future

the robust model with multimodal dataset will be used and the limitation of this study was that it was conducted on singular data on women and used limited models.

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